

Tufte-Oriented Revision Proposal

APA-UK007 After Storage and Shipment

Aim

Make the deck behave like evidence, not narration around evidence.

The revised presentation should let the audience see three things quickly:

- Chicago measured lower than the UK records.
- Current measured tensions remain inside the 4.0–8.5 N specification.
- U-layer conclusions are constrained by partial coverage and different access.

Tufte Principles Applied

Reduce

- Remove heavy theme treatment.
- Remove repeated legends.
- Keep only statistics used by the argument.
- Move background plots to backup.

Reveal

- Put comparable panels on shared scales.
- Label data directly.
- Show coverage limits visually.
- Tie each figure to one claim.

Current Deck Diagnosis

- The conclusion appears before enough visible evidence.
- Caveats are stated as bullets, not encoded in the figures.
- The audience must compare raw traces, then residual plots, then statistics.
- Long legends carry too much quantitative content.
- The dark inverted theme adds weight around sparse scientific plots.
- Absolute image paths make the deck less portable.

Revised Storyline

1. Finding and decision context.
2. Measurement basis and comparable coverage.
3. Primary residual evidence.
4. Current Chicago tensions relative to specification.
5. Wire-position structure.
6. What the data show and do not show.
7. Recommendation.
8. Backup for audit questions.

First Data Slide

Residual small multiples

Replace separate G and U change slides with one four-panel figure:

Panel	Data shown
G A	Chicago – UK by wire
G B	corrected B mapping residuals
U A	aligned partial subset
U B	aligned partial subset

Use shared y scales, faint points, a thin trend line, zero reference, and direct labels for n, median, mean, and percent lower in Chicago.

Why This Figure Leads

The central question is not the raw tension trace.

The central question is:

What changed between the UK record and the Chicago measurement?

Small multiples make the repeated pattern visible: four comparable layer-side groups all shift downward, while the magnitude and coverage differ.

Acceptance Figure

Current tensions versus specification

Add one figure focused on the acceptance question:

- One horizontal strip per layer-side group.
- Every current Chicago wire tension shown as a light point or rug mark.
- The 4.0–8.5 N spec shown as a pale band or two thin reference lines.
- Min, median, and max labeled once per group.

This separates historical change from current acceptability.

Coverage Figure

Make partial data visible

Add a compact coverage strip:

Row	Mark
G A	aligned UK and Chicago wires
G B	corrected aligned wires
U A	measured aligned subset
U B	measured aligned subset

Missing or unavailable regions should remain empty. The U-layer slit/access note belongs beside the affected rows, not on a separate caveat slide.

Raw Traces Move to Backup

Raw Chicago-versus-UK traces remain useful for auditability.

They should not carry the main argument unless the audience is inspecting a specific mapping or wire-region question.

Backup trace plots should use:

- one panel per layer-side group,
- shared y limits,
- thin marks,
- direct labels,
- no long statistical legends.

Plot Style Rules

- Use a white or very light background.
- Use color only for data roles.
- Remove chart borders and heavy grids.
- Put units in axis labels.
- Use direct labels instead of legends where practical.
- Use consistent axis ranges for intended comparisons.
- Prefer medians and quantile intervals for long-tailed measurements.
- Avoid count histograms when sample sizes differ.

Slide-Level Edits

- Replace `class: invert` with a light Marp theme.
- Replace absolute image links with relative paths.
- Fold G and U statistics into direct figure annotations.
- Move "Laser tension vs DWA" to backup unless needed live.
- Replace interpretation bullets with an evidence table.
- Make the recommendation distinguish change from acceptance.
- Move population profile-cloud plots to backup.

Proposed Deck Skeleton

1. APA-UK007 After Storage and Shipment
2. Measurement Basis and Comparable Coverage
3. Change in Tension, Chicago – UK
4. Current Chicago Tensions Relative to Specification
5. Is the Shift Spatially Structured?
6. What the Data Show and Do Not Show
7. Recommendation
8. Backup: Raw Chicago and UK Tension Traces
9. Backup: G and U Population Profiles

Regenerated Assets

Add new outputs in `dune_tension/experiments/UKAPA7_comparison`:

- `ukapa7_residual_small_multiples.png`
- `ukapa7_current_tension_vs_spec.png`
- `ukapa7_coverage_strip.png`

Extend

`src/dune_tension/ukapa7_comparison/generate_landscape_display_plots.py` so the deck remains reproducible from the source data.

Acceptance Check

The revised deck is ready when:

- the main result is visible from the first data figure,
- every numeric claim is next to the plot or table that supports it,
- the U-layer coverage limit appears with the U-layer result,
- current acceptance and historical change are visually separate,
- backup slides answer audit questions without competing with the main story.

Recommendation

Adopt the residual small-multiple figure as the core graphic, add a separate current-tension spec figure, and make measurement coverage explicit.

Then let the slides say less because the figures say more.